

This option parameter is only available if option 30.0 is non-zero.

This option parameter must be used when calibrating the oxygen probe.
There are three possible values:

Option parameter 30.6 value	Meaning
0	No calibrate. This is the normal operating condition, where the probe is used to measure the oxygen concentration in the flue.
1	Calibrate in air. The probe will be calibrated for 'offset' at the normal atmospheric oxygen concentration.
2	Calibrate in reference gas. The probe will be calibrated for 'gain' in a reference gas with a nominal oxygen concentration the level specified in option parameter 30.3 (typically 3.0%).

Once this calibration has been successfully completed the values in 30.1 and 30.2 will be automatically updated.

Option 30.7 - Boiler transport delay (5 – 60 seconds) LV3

Boiler transport delay is the time taken for 'gas' to travel from the burner to the oxygen probe. This delay varies with burner fire rate.

In order for the oxygen trim control loop to be stable, this parameter must be set accurately to the transport delay of the boiler **when at low fire**. This option may be set between **5** and **60** seconds inclusive.

- To measure the transport delay, ignite the burner and enter adjust ratio mode. Select the low fire setpoint and allow time for the flue oxygen reading to settle. Once the oxygen reading is steady, make a step change to the fuel/air ratio and start a timer. As soon as the measured oxygen reading **begins** to change, stop the timer. Set the option parameter to the recorded timer value in seconds.
- Automatic trim commissioning (option 30.9) will attempt to set this automatically. However the value must be checked by the engineer.

Option 30.8 - Reset oxygen trim profile (0/1) LV3

During full automatic operation, the O2 trim system will learn and store the percentage of trim applied to various firing rates. This provides a "feed forward" aspect to O2 trim. If any changes to target O2 values are made after the system has been in operation, the trim profile must be reset so as to not apply previous trim feed forward. A new trim profile will be learned as the burner modulates. Only the trim profile is affected. The Oxygen setpoints and flow values are not changed.

This option parameter allows the learned trim profile to be reset. If the option is set to 1 any learned profile currently held in memory will be reset and the trim drives will return to their commissioned positions for the corresponding modulation position. Once the profile has been reset this option parameter will automatically be reset to 0.

Option 30.9 – Automatic trim commissioning (0 / 1) LV3



The unit is able to automate the manual calculation procedures (option parameter 34.5), automatically calculating and entering flow values, oxygen trim setpoints, and boiler transport delay. The automatic commissioning procedure will only work if the following conditions are met:

- Option parameter 30.9 (automatic trim commissioning) is set to 1
- An oxygen probe is fitted and fully operational
- The unit is in adjust ratio with the burner firing a single fuel only
- A hydrocarbon ratio has been entered for the current fuel (see option parameters 35.1 — 35.4).

The auto commissioning procedure will usually take between 10 and 30 minutes (depending on number of setpoints), and is performed by the unit as follows:

1. Beginning with high fire, the unit moves the drives to each point in the firing range. **The display will show O2.**
2. When the measured oxygen reading settles, the unit stores the measured oxygen reading as the new oxygen setpoint.
3. The drives are moved to the next position directly below high fire (Ph-1) and wait for the oxygen to stabilize.
4. The unit moves the air drives up to the point above the current setpoint, leaving the fuel drive in the same position. **The display will show 'Flow calc An + 1'.**
5. When the new oxygen reading settles, the unit calculates and stores the new flow value. If the oxygen reading exceeds 15.0% during this stage, the unit shuts the burner down with F77.
6. When the unit has completed the low fire point, the measured boiler transport delay (at low fire) is stored in option parameter 30.7. Option parameter 30.5 (oxygen trim enable) is set to zero.
7. **IMPORTANT:** You must use "adjust ratio" mode to confirm each position from P3 (low fire) to Ph (high fire). If the flow values appear reasonable (about 4 to 6% change at each position) once you have finished, setting option 30.5 to 1 will enable O2 trim. There is a one-minute delay after exiting commissioning mode before trim will enable. If the flow values are unreasonable, they may be adjusted manually by placing the cursor to the "FLO" position and using the up/down arrows set the desired flow value. Using EK44 verify the O2 levels at each position and adjust to match if required. EK46 will help troubleshoot O2 trim issues.

BEFORE ENABLING TRIM, USE ADJUST RATIO MODE TO MANUALLY CHECK THE CALCULATED FLOW VALUES, OXYGEN SETPOINTS AND TRANSPORT DELAY. THE LOW FIRE PERCENTAGE FLOW VALUE SHOULD CONFORM APPROXIMATELY TO THE TURN-DOWN RATIO OF THE BURNER.

SUMMARY

To perform automatic trim commissioning, follow the procedure below:

1. If the probe is new, enter the offset and gain values in option parameters 30.1 and 30.2 which are supplied with the probe. To re-calibrate the probe refer to section 8.
2. Enter adjust ratio mode (See section 5).
3. Enable automatic trim commissioning by setting option parameter 30.9 to 1.
4. Wait for procedure to finish.
5. Check that the O₂, flow and transport delay values are sensible. Use EK44 for O2 value.
6. Check calibration of the oxygen probe.

7. Confirm setup using “adjust ratio” mode.

NOTE: If the boiler shuts off for any reason during automatic commissioning, the sequence will continue when the boiler re-starts.

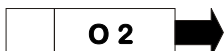
Option parameter 30.9 value	Meaning
0	Not selected.
1	Perform automatic trim characterization. The control will attempt to characterize the burner profile by calculating flow values and selecting oxygen setpoints to match the currently commissioned points. Automatic trim commissioning will be performed under the following conditions: • An oxygen probe is fitted and fully operational • The control is in adjust ratio mode. • A hydrocarbon ratio has been entered for the current fuel (see option parameters 35.1 – 35.4).

Option 31.0 - Limit Modulation Range (0 to 1) LV3

By default, the control modulates the burner between the set-points P_3 (low fire) and P_h (high fire), where P_h is the last set-point entered in commission ratio mode. If oxygen trim is fitted, it is not possible for any drive position to be trimmed lower than point P_3 (low fire) or higher than point P_h (high fire). Basically, you can't take fuel or air off at low fire, and you can't add fuel or air at high fire.

This option parameter allows the engineer to overcome this limitation in one of two ways. First the modulation range of the burner can be limited so that the burner may only modulate between points P_4 (one set-point above low fire) and P_{h-1} (one set-point below high fire). Since oxygen trim is not affected by this limitation, each drive may be trimmed between the points P_3 and P_h . This means that it is not possible to over-fire or under-fire the burner but a degree of trim may still be achieved at high and low fire. Note that this option may reduce the turn down of the boiler.

Option parameter 31.0 value	Meaning
0	Modulation not limited. Burner modulates from P_3 to P_h. Oxygen trim works over full range from P_3 to P_h, but may not be able to apply trim at high fire, or low fire – depending on trim direction. The burner always modulates to the desired modulation rate, which may limit the system's ability to maintain the desired oxygen value.
1	Normal modulation range limited to P_4 to P_{h-1} always. Oxygen trim works over full range from P_3 to P_h, and can apply extra trim at low and high fire – limited by P_3 and P_h. This setting is useful when there is extra capacity (usually air flow) available for the burner, but otherwise it will reduce the turn-down ratio of the boiler.



Options 31.1 to 31.4 - Trim Type for profiles 1 to 4 (0 to 2) LV3

These option parameters determine which drive will be trimmed (Air or Fuel) in profiles 1 to 4. These parameters are used together with the parameters entered in option parameters 2.0 to 2.9 to determine which drives are trimmed, and in which direction.

Parameter	Meaning
31.1	Trim Type for profile 1
31.2	Trim Type for profile 2
31.3	Trim Type for profile 3
31.4	Trim Type for profile 4

Parameter value in 31.X	Meaning
0	No Trim. Do not apply oxygen trim when running on this profile.
1	Air Trim. Trim 'Air' drives when running on this profile (see option parameters 2.0 to 2.9.)
2	Fuel Trim. Trim 'Fuel' drives when running on this profile (see option parameters 2.0 to 2.9.)

Option 32.0 - Trim limit default (0/1) LV3

0 = Default Trim Limit of $\pm 5\%$
1 = Enables options 32.1 to 32.4

Options 32.1 to 32.4 - Trim limits (0.0 – 25.0) LV3

Option 32.0 is only available if option 30.5 (oxygen input function) has been set for closed loop oxygen trim. It is possible for the system to trim the air or fuel drives up to a maximum deviation of $\pm 25.0\%$ of the total air or fuel flow for the chosen trim drives.

If option 32.0 is set to 0, the default trim limit will be $\pm 5.0\%$ for all profile selections. Option parameters 32.1 to 32.4 will not be available.

If option 32.0 is set to 1, it is possible to individually adjust the trim limit for each profile selection using option parameters 32.1 to 32.4. The trim limit may be adjusted from 0.0% (no trim) up to a maximum of $\pm 25.0\%$ (maximum allowable trim).

It is the responsibility of the commissioning engineer to ensure the trim limit set will not allow a hazardous combustion condition to occur in the event of an oxygen probe failure. Where oxygen is to be considered safety critical, option parameters 31.X should be used to setup a second oxygen monitor. When a second oxygen monitoring system is used, the oxygen can be considered fail-safe, and the trim limits can be set up to 50% instead of 25%.

Options 33.1 to 33.4 - Trim integral gain (Default 10%) LV3

Options 33.1 to 33.4 are only available if option 30.5 (oxygen input function) has been set for closed loop oxygen trim.



In order for the oxygen trim control loop to be stable, the integral gain must be set correctly. Options 33.1 to 33.4 allow the integral gain to be set individually for each profile combination to any value between 0.0 and 99.9%. As a general recommendation, the integral gain should be initially set to 10.0%. This value must be set above 0.0 for the system to apply trim.

If the burner is firing, it is only possible to make an adjustment to the option parameter that relates to the selected profile. If the burner is not firing, the integral gain for any profile selection may be adjusted.

Option 34.0 - Trim proportional gain (0/1) LV3

Options 34.1 to 34.4 - Trim proportional gain (0.0 – 99.9%) LV3

Options 34.0 to 34.4 are only available if option 30.5 (oxygen input function) has been set for closed loop oxygen trim.

If oxygen trim proportional gain is not required for any profile selection, set option parameter 34.0 to **0**.

If oxygen trim proportional gain is required, set option parameter 34.0 to **1**. Option parameters 34.1 to 34.4 will then become available. Options 34.1 to 34.4 allow the loop gain to be set individually for each profile selection to any value between 0.0 and 99.9%.

Oxygen trim proportional gain is not normally required.

Option 34.5 - Calculating and entering the flow values manually LV3

If the flow values are to be calculated manually, the procedure below must be followed.

Enter Adjust Ratio Mode with oxygen trim **disabled** as outlined above.

1. Select the high fire position. This will cause the display to show A(n), where n is the number of the high fire profile point.
2. Wait until the oxygen reading has stabilized, the value can be viewed using the engineer's key.
3. Record the **excess air** value at location x in the table. For single fuel profiles, this value may be found using the Engineer's Key parameter EK48 (provided the hydrocarbon ratio has been entered into the appropriate option parameter (35.1 - 35.4) for this fuel). For multiple fuel profiles, **Refer to NOTE under section Options 35.1 to 35.4.**
4. Select the profile position immediately below high fire. This will cause the display to show A(n - 1).
5. Wait until the oxygen reading has stabilized and record the **excess air** value at location a₁ in the table
6. Move air dampers 1 and 2 (if fitted) **up** until they stop, so that they are at the setpoint immediately above the current position. **Do not press enter.**
7. Wait until the oxygen reading has stabilized, then record the **excess air** value at location b₁ in the table, relating to the current profile position

8. Repeat the above for all other profile positions including low fire (profile position A3), recording each time the values at locations a and b in the table. When extra air is added at P3, the time taken before the flue oxygen reading starts to increase should be measured and entered into option parameter 30.7.
9. After completing the table for all Excess Air values a and b, complete the Excess Air + 100 column, by adding 100 (i.e. $y = x + 100$, $c = a + 100$ and $d = b + 100$).
10. Complete the Ratio column by dividing c by d (i.e. $e = c/d$).
11. Complete the Airflow column by multiplying e by the previous value of f (i.e. $f_x = e_x f_{x-1}$).
12. If the system will be applying trim to the fuel, the fuel flow column must be completed. This is achieved by multiplying the airflow at each point by the ratio of excess air + 100 at high fire divided by the excess air + 100 at the actual point (i.e. $g_x = f_x(y/c_x)$).

Example of table completed for A10 (high fire) to A8. In practice, the table must be filled out down to A3 (low fire).

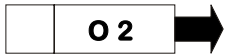
Profile position	Excess Air		Ex. Air + 100 $y = x + 100$ $c = a + 100$ $d = b + 100$		Ratio	Airflow	Fuel flow
High fire = A10	x	10	y	110	-----	99.9%	99.9%
1 = A9	a ₁	8	c ₁	108	$e_1 = c_1/d_1$	$f_1 = e_1 (99.9)$	$g_1 = f_1 (y/c_1)$
	b ₁	33	d ₁	133	0.812	81.1%	82.6%
2 = A8	a ₂	9	c ₂	109	$e_2 = c_2/d_2$	$f_2 = e_2 (f_1)$	$g_2 = f_2 (y/c_2)$
	b ₂	20	d ₂	120	0.908	73.7%	74.4%



Table: Flow calculation table

Profile position	Excess Air		Ex. Air + 100 $y = x + 100$ $c = a + 100$ $d = b + 100$		Ratio	Airflow	Fuel flow
High fire = A	x		y				
1 = A	a ₁ b ₁		c ₁ d ₁		$e_1 = c_1/d_1$	$f_1 = e_1 (99.9)$	$g_1 = f_1 (y/c_1)$
2 = A	a ₂ b ₂		c ₂ d ₂		$e_2 = c_2/d_2$	$f_2 = e_2 (f_1)$	$g_2 = f_2 (y/c_2)$
3 = A	a ₃ b ₃		c ₃ d ₃		$e_3 = c_3/d_3$	$f_3 = e_3 (f_2)$	$g_3 = f_3 (y/c_3)$
4 = A	a ₄ b ₄		c ₄ d ₄		$e_4 = c_4/d_4$	$f_4 = e_4 (f_3)$	$g_4 = f_4 (y/c_4)$
5 = A	a ₅ b ₅		c ₅ d ₅		$e_5 = c_5/d_5$	$f_5 = e_5 (f_4)$	$g_5 = f_5 (y/c_5)$
6 = A	a ₆ b ₆		c ₆ d ₆		$e_6 = c_6/d_6$	$f_6 = e_6 (f_5)$	$g_6 = f_6 (y/c_6)$
7 = A	a ₇ b ₇		c ₇ d ₇		$e_7 = c_7/d_7$	$f_7 = e_7 (f_6)$	$g_7 = f_7 (y/c_7)$
8 = A	a ₈ b ₈		c ₈ d ₈		$e_8 = c_8/d_8$	$f_8 = e_8 (f_7)$	$g_8 = f_8 (y/c_8)$
9 = A	a ₉ b ₉		c ₉ d ₉		$e_9 = c_9/d_9$	$f_9 = e_9 (f_8)$	$g_9 = f_9 (y/c_9)$
10 = A	a ₁₀ b ₁₀		c ₁₀ d ₁₀		$e_{10} = c_{10}/d_{10}$	$f_{10} = e_{10} (f_9)$	$g_{10} = f_{10} (y/c_{10})$
11 = A	a ₁₁ b ₁₁		c ₁₁ d ₁₁		$e_{11} = c_{11}/d_{11}$	$f_{11} = e_{11} (f_{10})$	$g_{11} = f_{11} (y/c_{11})$
12 = A	a ₁₂ b ₁₂		c ₁₂ d ₁₂		$e_{12} = c_{12}/d_{12}$	$f_{12} = e_{12} (f_{11})$	$g_{12} = f_{12} (y/c_{12})$
13 = A	a ₁₃ b ₁₃		c ₁₃ d ₁₃		$e_{13} = c_{13}/d_{13}$	$f_{13} = e_{13} (f_{12})$	$g_{13} = f_{13} (y/c_{13})$
14 = A	a ₁₄ b ₁₄		c ₁₄ d ₁₄		$e_{14} = c_{14}/d_{14}$	$f_{14} = e_{14} (f_{13})$	$g_{14} = f_{14} (y/c_{14})$
15 = A	a ₁₅ b ₁₅		c ₁₅ d ₁₅		$e_{15} = c_{15}/d_{15}$	$f_{15} = e_{15} (f_{14})/15$	$g_{15} = f_{15} (y/c_{15})$
16 = A	a ₁₆ b ₁₆		c ₁₆ d ₁₆		$e_{16} = c_{16}/d_{16}$	$f_{16} = e_{16} (f_{15})$	$g_{16} = f_{16} (y/c_{16})$
17 = A	a ₁₇ b ₁₇		c ₁₇ d ₁₇		$e_{17} = c_{17}/d_{17}$	$f_{17} = e_{17} (f_{16})$	$g_{17} = f_{17} (y/c_{17})$
18 = A	a ₁₈ b ₁₈		c ₁₈ d ₁₈		$e_{18} = c_{18}/d_{18}$	$f_{18} = e_{18} (f_{17})$	$g_{18} = f_{18} (y/c_{18})$
19 = A	a ₁₉ b ₁₉		c ₁₉ d ₁₉		$e_{19} = c_{19}/d_{19}$	$f_{19} = e_{19} (f_{18})$	$g_{19} = f_{19} (y/c_{19})$
20 = A	a ₂₀ b ₂₀		c ₂₀ d ₂₀		$E_{20} = c_{20}/d_{20}$	$f_{20} = e_{20} (f_{19})$	$g_{20} = f_{20} (y/c_{20})$

NOTE: For the following Options 35.x, see procedure in Section 9.5.



Option 35.0 – Inlet temperature sensor serial number LV3

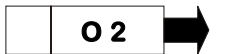
Each CANbus device connected to the system has a unique serial number allocated to it during production testing. This number cannot be changed and is used to uniquely identify each device.

When the system is in commissioning mode the serial number of all the connected devices is read by the PPC6000 and displayed as a 'list' to allow the selection of the relevant serial number unit to be made.

Scrolling the list in an upward direction will scroll through all temperature sensor options available until '0, nothing' is displayed. Enter this value to disconnect the specified device from any hardware.

Scrolling the list in a downward direction will stop when the currently selected item is displayed, and the value will stop flashing.

The display will show type information for each device present (such as 'air temp'), but the commissioning engineer should be aware of which serial number device is being used for which function.

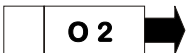


Options 35.1 to 35.4-Hydrocarbon ratios of each fuel respectively (0.00 – 9.99) LV3

If a display of calculated burner efficiency, or 'automatic trim commissioning' is required, the hydrocarbon ratios for the required fuels must be entered. These may be set between **0.0** and **9.99** inclusive, where the value entered is the hydrocarbon ratio x 10. For example, for a hydrocarbon ratio of 0.157, a value of **1.57** should be entered.

Note: When firing multiple fuels, the mixture between the fuels fired may vary across the firing range of the burner. When this happens, the effective hydrocarbon ratio and calorific value of the fuel combination will change. The accuracy of the calculated efficiency and the 'automatic trim commissioning' procedure will be adversely affected by this. It maybe that these functions should not be used, in which case these values should be entered as zero.

See Option Table 35.x A on page 119.



Options 35.5 to 35.8-Calorific values of fuels/profiles 1 to 4 respectively (0.0 – 99.9) LV3

These option parameters are only available if option 35.0 (inlet temperature sensor) is not set to zero.

If a display of calculated burner efficiency is required, the calorific value of the required fuels must be entered. May be set between **0.0** and **99.9 MJkg⁻¹** inclusive.

Note: When firing multiple fuels, the mixture between the fuels fired may vary across the firing range of the burner. When this happens, the effective hydrocarbon ratio and calorific value of the fuel combination will change. The accuracy of the calculated efficiency and the 'automatic trim commissioning' procedure will be adversely affected by this. It maybe that these functions should not be used, in which case these values should be entered as zero.

The following table shows the calorific values and hydrocarbon ratios of several common fuels. These values should be used as a guide only. If the display is required to show gross efficiency in addition to entering the gross calorific values a non zero value for the boiler radiated heat loss **MUST** be entered.

See Option Table 35.x A on page 119.

Option Table 35.x A

Fuel	Gross calorific value	Net calorific value	Hydrocarbon ratio (x10)
Gas	52.8	47.6	3.20
Light oil	45.6	42.8	1.57
Medium oil	43.1	40.8	1.35
Heavy oil	42.9	40.5	1.28

0 2 → Option 35.9 - Boiler radiated heat loss (0.0 – 9.9%) LV3

This option parameter is only available if option 35.0 (inlet temperature sensor) is not set to zero.

If this option parameter is set to zero the unit will calculate the net efficiency, if non-zero it will calculate gross efficiency.

For accurate calculation of boiler gross efficiency, this option parameter must be set to the value of the heat lost through the shell of the boiler at high fire as a percentage of the burner's output at high fire. The range of this parameter is **0.0** to **9.9%**.

Option 36.0 - Flue temperature alarm select (0/1) LV3

This option parameter is only available if option 30.0 is non zero.

If option 36.0 is set to **0**, the flue temperature high and low alarms are disabled.

If option 36.0 is set to **1**, the flue temperature high and low alarms are enabled and option parameters 36.1 to 36.4 and 37.1 to 37.4 become available.

Options 36.1 to 36.4 - Flue temperature low alarm values (0 – 999) LV3

Using option parameters 36.1 to 36.4, it is possible to set a different flue temperature low alarm value for each profile combination. Each option parameter may be set to any value between **0** and **999** inclusive. The temperature units should be set with respect to option parameter 30.4.

If the burner is firing, it is only possible to make an adjustment to the option parameter that relates to the selected profile. If the burner is not firing, the low alarm value for any profile selection may be adjusted.

When the flue temperature drops below the low alarm value for the selected profile, a fault number will appear.

Options 37.1 to 37.4 - Flue temperature high alarm values (0 – 999) LV3

Using option parameters 37.1 to 37.4, it is possible to set a different flue temperature high alarm value for each profile selection. Each option parameter may be set to any value between **0** and **999** inclusive. The temperature units should be set with respect to option parameter 30.4.

If the burner is firing, it is only possible to make an adjustment to the option parameter that relates to the selected profile. If the burner is not firing, the high alarm value for any profile selection may be adjusted.

When the flue temperature rises above the high alarm value for the selected profile combination, a fault number will appear.

O 2 → Option 38.0 - Oxygen alarm select (0 – 2) LV3



CAUTION

This parameter is not required, but *highly recommended*.

Option parameter 38.0 is only available when option 30.5 (oxygen input select) is not set to zero and the optional oxygen probe interface unit is connected.

Since the oxygen alarm values relate to the oxygen setpoint values, oxygen setpoint values must first be entered in adjust ratio mode.

Option parameter 38.0 value	Meaning
0	Alarms disabled. Oxygen low and high alarms are disabled.
1	Alarms enabled, no non-volatile lockout. Oxygen low and high alarms are enabled and option parameters 38.1 to 41.4 will become available. If an alarm value is exceeded, a fault number will appear but no non-volatile lockout will occur.
2	Alarms enabled with non-volatile lockout. Oxygen low and high alarms are enabled and option parameters 38.1 to 41.4 will become available. If an alarm value is exceeded, a fault number will appear and a non-volatile lockout will occur.

O 2 → Options 38.1 to 38.4 - Oxygen low alarm values at low fire (0.0 – 99.9%) LV3

O 2 → Options 39.1 to 39.4 - Oxygen low alarm values at high fire (0.0 – 99.9%) LV3

O 2 → Options 40.1 to 40.4 - Oxygen high alarm values at low fire (0.0 – 99.9%) LV3

O 2 → Options 41.1 to 41.4 - Oxygen high alarm values at high fire (0.0 – 99.9%) LV3

Using option parameters 38.1 to 41.4, it is possible to set different oxygen alarm values at high fire and low fire for each profile.

At firing positions other than high and low fire, the alarm limit is interpolated between the two points. For example, if the oxygen limit has been entered as 5% of setpoint at low fire and 10% of setpoint at high fire, the limit at mid fire will be 7.5%.